

How reproducible are AI-ECG outputs between ECGs recorded hours apart?

Age and recorded sex cannot change within hours, so any difference between two ECGs is variability

Design

3 cohorts

HEEDB (MGH, Emory),
MIMIC-IV-ECG

532 891 pairs

consecutive ECGs,
253 995 patients

8 intervals

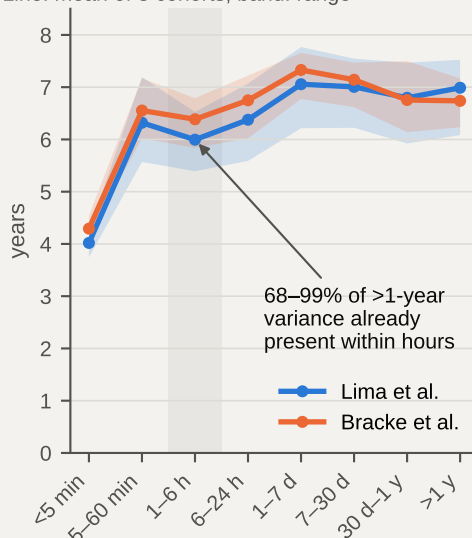
<5 min to >1 year

Models

2 published ECG-age
1 published diagnostic
in-house age & sex

ECG-age gap: within-patient SD

Line: mean of 3 cohorts; band: range



Over 1-6 hours

±15-19 y

smallest detectable change
in ECG age (published models)

18-19%

change "ECG age >8 y older"
category

13-15%

change predicted sex

13-22%

of bundle branch block calls
lost (machine reads no better)

Take-home

A single AI-ECG estimate is not a precise individual measurement. Report repeatability with accuracy, average repeated ECGs, and use model ensembles before interpreting serial change in ECG age.